

Eigenvalues and inverse problems

The optimal decay exponent for the conditioning of sign matrices

Difficulty: extreme **Importance:** interesting to the community

Status: Open

Rating rationale: Extreme reflects a sharp asymptotic rate tied to longstanding gaps in Hadamard constructions; community impact is a quantitative limit on nearly orthogonal sign matrices.

Context and notation

For $n \geq 1$, define

$$h(n) = \min_{A \in \{-1,1\}^{n \times n}} \kappa_2(A), \quad \kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)},$$

with $\kappa_2(A) = +\infty$ for singular A .

Problem statement

Determine the exact number

$$\alpha_* = \sup\{\alpha \geq 0 : \exists C > 0 \forall n \geq 1, h(n) - 1 \leq Cn^{-\alpha}\}.$$

The constant C may depend on α , but not on dimension. This specifies the uniform asymptotic power exponent; logarithmic factors do not change the supremum. Orders admitting exact Hadamard matrices have $h(n) - 1 = 0$, which are included without taking logarithms of zero. Current bounds give $17/92 \leq \alpha_* \leq 1$.

References

Alexeev, Jasper, and Mixon, *Asymptotically optimal approximate Hadamard matrices*, §6, Problem 11; the supremum above is an editorial precise formulation of its decay-exponent question. Steinerberger, *Open Problems*, Problem 69, November 2025 update, identifies the sharp rate as unresolved.

Earlier status check — 2026-09-08

Searches for *approximate Hadamard 17/92 2026* and *approximate Hadamard condition 2026 sharp* found no exact exponent. A global constant as in [IS-04](#) does not specify a decay exponent, while an asymptotic exponent permits finitely many exceptions to any proposed sharp constant. The two independently posed problems are therefore retained separately.

Audit update — 2026-09-10

Rechecked [Alexeev–Jasper–Mixon](#), §6, Problem 11, and [Steinerberger’s problem collection](#), Problem 69. Both leave the sharp rate unresolved. Searches for later approximate-Hadamard conditioning exponents found no determination of the optimum; a nonsharp exponent bound does not by itself resolve the target.